



Figure 3. Ring network of Example 1.

**Example 1.** Without loss of generality, we fix an article that has message  $m = R$  and some intermediate reliability  $r$ , so that ex-ante the article may or may not contain misinformation. We also condition on an article that turns out to contain misinformation. Moreover, we assume the regulator has only a semi-effective provenance policy ( $\rho = 1/4$ ) and the sharing network is a ring, as pictured in Figure 3, so that virality is simply the expected distance between the seed agent and the first agent who elects not to share.

We also assume that strategic complementarities play a small role (i.e.,  $\kappa \approx 0$  and  $d \approx 0$ ), so sharing decisions are mainly based on an agent's assessment of the article's veracity. Moreover, we assume all prior beliefs are drawn from the same distribution  $H$ . First, we consider the virality of content with no provenance policy. Per Theorem 1, there exists some  $b^{**}$  such that agents with  $b_i > b^{**}$  share and agents with  $b_i < b^{**}$  do not share. Hence, each agent shares the article with probability  $\zeta \equiv 1 - H(b^{**})$ , and the expected spread of the article is given by  $\sum_{i=1}^N \zeta^i$ . Note that this probability applies regardless of whether the article contains misinformation, which is not observable to the agent without fact-checking.

Next, consider the introduction of a provenance policy with  $\rho = 1/4$ , so the likelihood that an agent fact-checks the article before choosing her action is  $1/4$  and the likelihood that an agent takes her action without fact-checking is  $3/4$ . Because, by assumption, the article contains misinformation, a fact-check will set agent  $i$ 's belief of veracity to zero, ensuring that she does not share this article. Therefore, the overall probability that agent 1 will share the article is  $(1 - \rho)\zeta = 3\zeta/4$ . While a fact-check immediately kills the article (which occurs with probability  $1/4$ ), no fact-check leads to sharing with probability  $\zeta$  (also occurring with probability  $3/4$ ).

Given this provenance policy, we now consider agent 2, who next receives the article. Agent 2 does not know whether the article has already been fact-checked by agent 1, but knows the provenance policy is in place. Given the article was passed to her, she assesses that there is a higher probability that the article is truthful relative to the case with no provenance policy. This is because the act of agent 1 sharing the article with agent 2 may have been the result of a fact-check that came back positive (article is truthful), and so observing this action leads her to a higher posterior belief the article is truthful. Consequently, the prior cutoff for sharing will be below  $b^{**}$  (the cutoff without provenance), and if she does not fact-check herself, her likelihood of sharing will be some  $\zeta_2 > \zeta \equiv \zeta_1$ . As before, the probability that agent 2 will share is  $(1 - \rho)\zeta_2 = 3\zeta_2/4$ . Extending this reasoning, the probability agent  $i$  will share is  $(1 - \rho)\zeta_i$ , where  $\zeta_i > \zeta_{i-1}$  for all  $i$ .

Thus, to determine whether the provenance policy is effective, one needs to compare  $\sum_{i=1}^N \zeta^i$  and